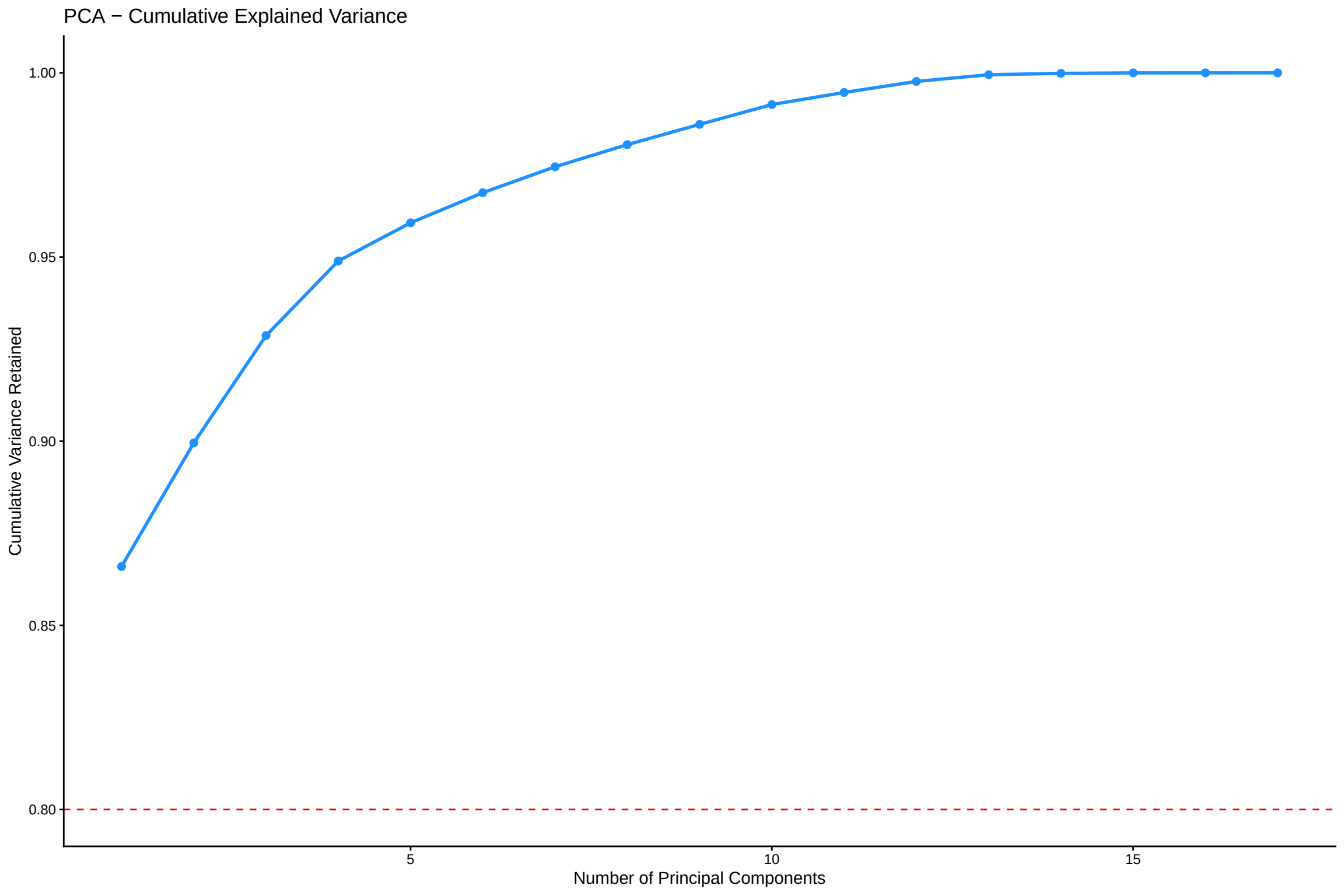
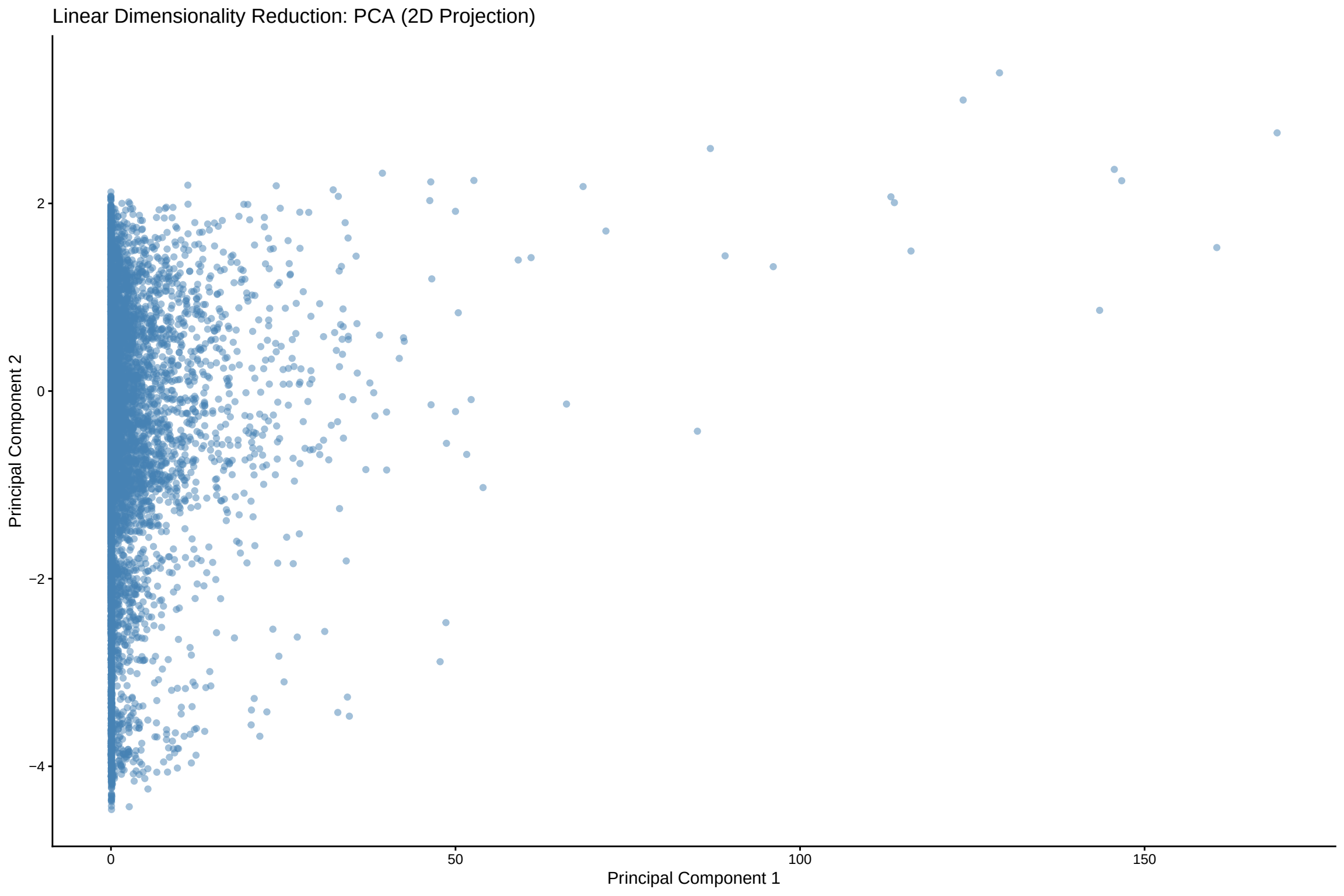
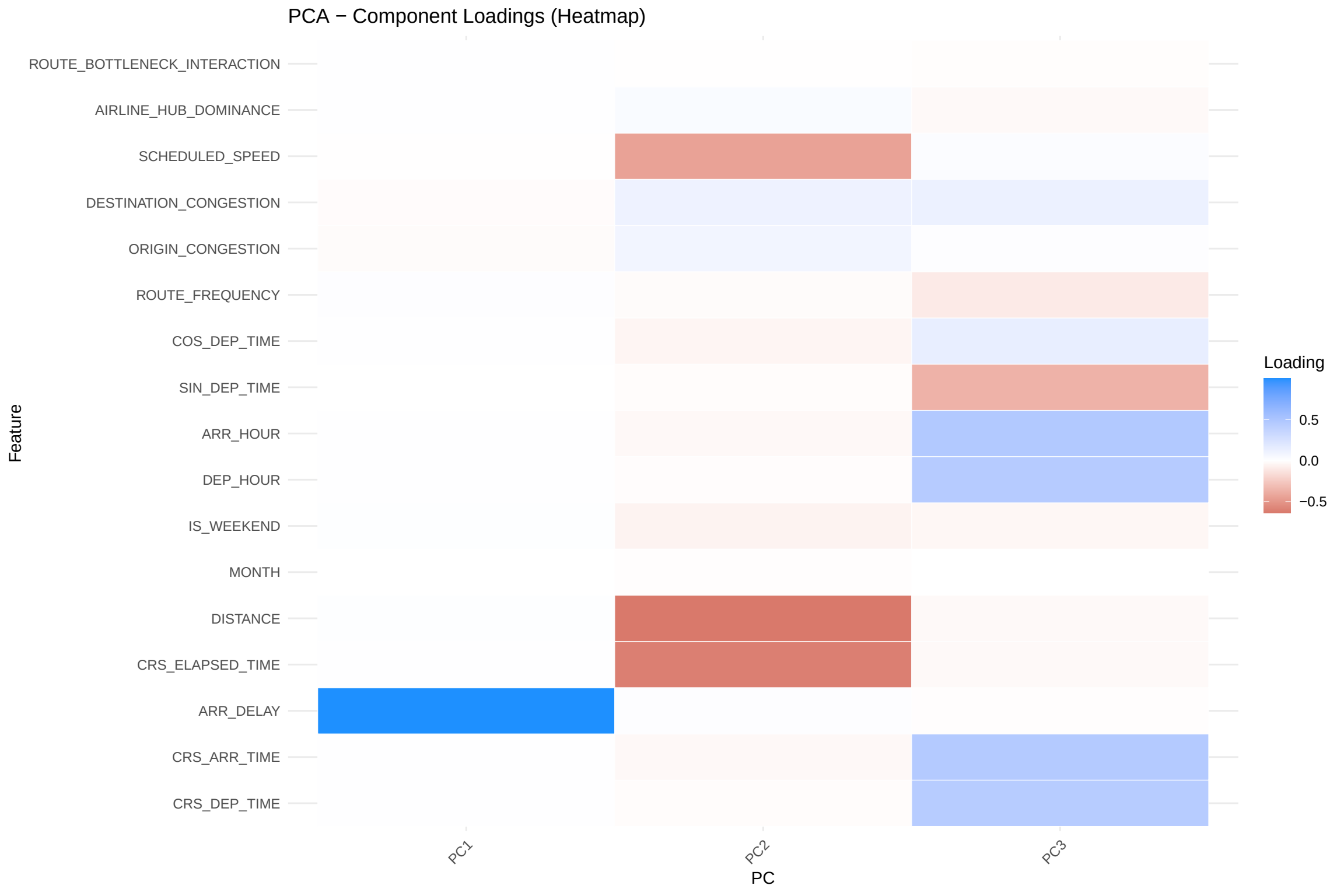
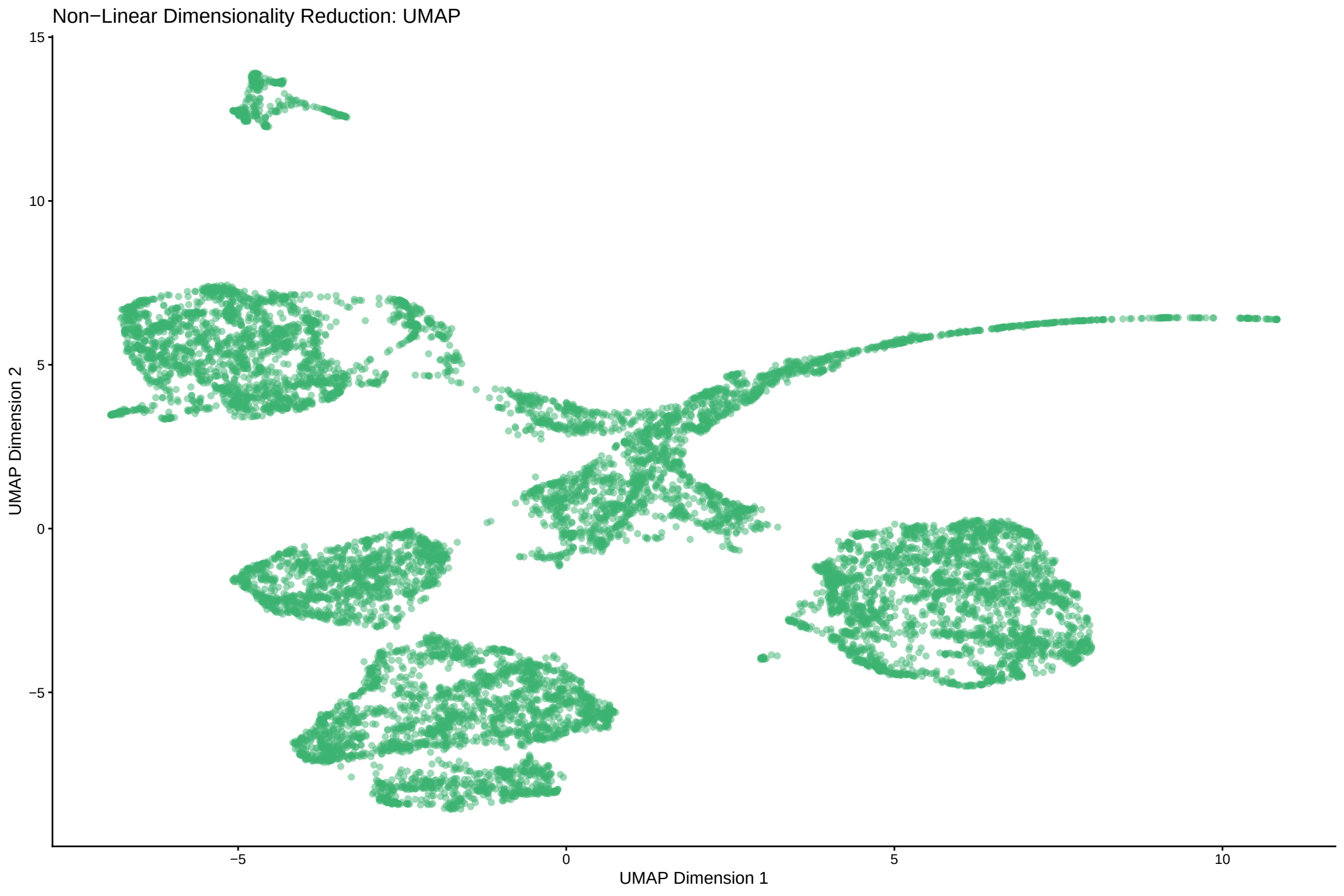

PCA − Cumulative Explained Variance
Cumulative Variance Retained
1.00
0.95
0.90
0.85
0.80
5
10
15
Number of Principal Components


Linear Dimensionality Reduction: PCA (2D Projection)
Principal Component 2
2
0
−2
−4
0
50
100
150
Principal Component 1


PCA − Component Loadings (Heatmap)
ROUTE_BOTTLENECK_INTERACTION
AIRLINE_HUB_DOMINANCE
SCHEDULED_SPEED
DESTINATION_CONGESTION
ORIGIN_CONGESTION
ROUTE_FREQUENCY
COS_DEP_TIME
SIN_DEP_TIME
ARR_HOUR
DEP_HOUR
IS_WEEKEND
MONTH
DISTANCE
CRS_ELAPSED_TIME
ARR_DELAY
CRS_ARR_TIME
CRS_DEP_TIME
Feature
PC1
PC2
PC3
PC
Loading
0.5
0.0
−0.5


Non−Linear Dimensionality Reduction: UMAP
15
10
5
0
−5
UMAP Dimension 2
−5
0
5
10
UMAP Dimension 1